MIL-F-5003C(ME)
18 October 1973
SUPERSEDING
MIL-F-5003B(MO)
13 January 1965

MILITARY SPECIFICATION

FLOODLIGHT SET, ELECTRIC, TRIPOD-MOUNTED, FOUR 150-WATT PROJECTOR LAMPS

1. SCOPE

1.1 This specification covers a floodlight set, electric, tripod-mounted (hereinafter referred to as lamp assembly) designed for use in areas where explosion proof type illumination equipment is not required (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-C-596	- Connector, Plug, Electrical; Connector, Receptacle Electrical.
W-L-661	- Light, Extension, Electrical.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements for.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6230

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Military

MIL-T-152	- Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-P-6808	- Primer Coating, Zinc Chromate, for Aircraft and Missile Applications; Application of.
MIL-C-12044	- Chests, Plywood.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AIR POLLUTION CONTROL DISTRICT

Rule 66

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001.)

3. REQUIREMENTS

3.1 Description. The lamp assembly shall be in accordance with figures 1 through 5 and as otherwise specified herein. Dimensions and tolerances are shown where required. The supplier is responsible for preparing his own drawings. Where tolerances are not shown, the supplier is responsible for providing tolerances to insure proper fit, assembly, and operation of each assembly. No deviation from dimensions or tolerances on the figures is permissible without approval of the contracting officer.

3.2 First article (first produced lamp assembly). The supplier shall furnish the first two lamp assemblies produced from production tooling, for examination and testing within the time frame specified (see 6.2) to prove that his methods will produce lamp assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Operation. The lamp assembly connected to a 120-volt, ac source, the lamp cords and extension light plugged into the outlet box, and the switches in the "on" position shall energize all the lamps. The lamp assembly shall show no signs of electrical or mechanical malfunction or damage as a result of the stand being erected to its maximum position and retracted to its carrying position.

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3.4 Materials. Metals shall be of the corrosion-resistant type or treated to resist corrosion. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other.

3.5 Electrical cable and connectors. The electrical cable and connectors shall conform to MIL-C-3432 as specified for low-temperature, heat-resistant cable (ISL for insulation and JSL for jacket). The connectors, plugs, and receptacles shall conform to W-C-596. The connectors shall be molded and shall be vulcanized to the cable.

3.6 Lamp-head assembly. The lamp-head assembly (see figure 1) shall be provided with a positive means to lock the assembly to the mast. The lamp-head housing cover shall be removable to provide access to the switch and wiring. All electrical wiring shall be insulated from the metal components of the housing. The electrical cable shall be 2-conductor, No. 18 A.W.G., light-duty 12-1/2 feet plus 1/2 minus 0 feet long, and terminated on one end with a molded connector equipped with two parallel-contact blades for connecting to the electrical connector box. The electrical switch for turning the lamps on and off shall be a flush-mounted, weatherproof type.

3.7 Stand. The stand shall consist of a base, a mast, and a crossarm.

3.7.1 Base. The base shall conform to figures 2 and 3 and shall have legs that can be folded as shown in figure 4 for shipment or transport and locked in place during use. A strap shall be fastened to the crossarm for holding the legs and crossarms when in the retracted position.

3.7.1.1 Pegs and casters. Three-inch pegs and casters with step-on wheel brakes shall be provided to support the stand assembly. They shall be interchangeable.

3.7.2 Mast and legs. All exterior surfaces of the welded tubing shall be cadmium plated in accordance with QQ-P-416 class 1, type II. Interior surfaces of the tubing shall be coated with zinc chromate primer in accordance with MIL-P-6808.

3.7.2.1 Stops and clamping screws. Telescoping sections shall be provided with stops for preventing section from being detached from their mating tubes. T-type hand screws and shoes, or other comparable means, shall be provided for clamping the telescopic sections in position.

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3.7.3 Crossarms. The crossarms shall be constructed to allow selective elevating and clamping of the arm along the lower section of the mast, and the arms shall be held in place by the same type of clamp that is used on the telescoping sections. Crossarms shall be fabricated to permit folding downward and parallel to the main tube for shipment and locking in place during use. One swivel lampholder shall be mounted on each crossarm. Each lampholder shall be equipped with 10 feet plus 1/2 minus 0 feet of 2-conductor, No. 18, light-duty, electrical cable terminated at one end with a molded connector. The connector shall have two parallel contact blades for connecting to the electrical connector box.

3.8 Swivel lampholder. The swivel lampholder shall be weatherproof and equipped with a wire guard as shown on figure 5 or a sheet metal guard with a hinged wire lamp guard over the open end. Electrical cord length shall be as specified in 3.6 and 3.7.3. The socket in the lampholder shall not fail when a 60 inch-pound torque is applied to the lamp installed in the socket. The following information shall be permanently and legibly marked on the lamp-head adjacent to the switch: "CAUTION-TURN OFF BEFORE REPLACING LAMPS. USE PAR-38 LAMP WITH MEDIUM SCREW SKIRTED BASE."

3.9 Lamps. Lamps shall be furnished by the supplier as part of the complete assembly, and an equal number of replacement lamps shall also be furnished. The portable carrying case (see 3.12) shall contain eight 120-V, 150-W, medium screw skirted base, PAR-38 projector floodlamps and two standard 120-V, 100-W lamps for extension light.

3.10 Electrical connector box. The electrical connector box shall be a weatherproof type containing six weatherproof outlets and two weatherproof switches. One switch shall control the first four outlets which will receive the connector plugs from the swivel lampholders. The other switch shall control two outlets of the 3-prong, polarized, twist-lock type for use with the extension light. The connector box shall be provided with 100 feet plus 1/2 minus 0 feet of No. 12 medium-duty, 3-conductor electrical cable terminated at one end with a polarized, heavy-duty, outdoor-service-type connector plug having two parallel contacts and one ground connector contact. The third conductor shall be grounded to the mast.

3.11 Extension light. One portable extension light equipped with 50 feet of No. 18 medium-duty, 3-conductor electrical cable of the type specified in 3.5 shall be furnished. The portable extension shall be in accordance with W-L-661, type II, class 2, except that it shall include a switch. The cable shall be terminated at one end with a 3-prong, polarized, twist-lock, molded connector plug in accordance with W-C-596.

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3.12 Carrying case. The portable carrying case shall be constructed in accordance with MIL-C-12044 of 3/8-inch-thick plywood. A hinged top shall be secured to the sides of the carrying case by means of snap fasteners. A chain shall be provided to limit the top from opening more than 100 degrees. Hand supports shall be provided on each end of the portable case. Each lamp assembly, complete with replacement lamps, shall be packed in a carrying case. Each component, part, or subassembly shall be secured within the carrying case to prevent movement and breakage during handling and shipping. An 8-by 10-inch loading plan shall be permanently attached to the inside cover of the carrying case.

3.13 Treatment and painting. After installation, circuit elements and wiring shall be treated as specified in MIL-T-152 with a varnish conforming to MIL-V-173 composition as specified in 3.13.1. All ferrous metal parts normally painted shall be cadmium plated in accordance with QQ-P-416, class 1, type II, or zinc plated in accordance with QQ-Z-325, type II, class 2, prior to painting. The case shall be treated and painted in accordance with MIL-C-12044. The metal parts and words "CARRYING CASE" shall be painted in accordance with MIL-T-704, except that the finish color shall be as specified (see 6.2).

3.13.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.14 Operational reliability. The lamp assembly shall withstand without malfunction, damage or permanent deformation 100 operational cycles when tested as specified in 4.6.2.

3.15 Identification marking. Assemblies and parts shall be marked in accordance with MIL-STD-130.

3.16 Workmanship. The lamp assembly shall be free from defects such as chips, burrs, loose electrical connections, bent tubing and misalined components.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any test by the first produced lamp assemblies shall be cause for disassembly, in the presence of a Government representative, of the first produced lamp assemblies to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced figures in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First produced lamp assembly inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.7).

4.3 First produced lamp assembly inspection.

4.3.1 Examination. Each first produced lamp assembly shall be examined in accordance with table I. Presence of one or more defects in either lamp assembly shall be cause for rejection of both lamp assemblies.

4.3.2 Tests. Following successful completion of the examination, the first produced lamp assemblies shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for performing the inspection specified in 4.1.1.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S2. AQL shall be 6.5 percent defective.

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4.4.2 Examination. Lamp assemblies selected in accordance with 4.4.1 shall be examined in accordance with table I and compared with the first produced lamp assemblies. Each lamp assembly having one or more defects or which deviates from the first produced lamp assembly shall be considered defective and shall be cause for rejection of the lot.

4.4.3 Tests.

4.4.3.1 Individual. Each lamp assembly which has passed the examination specified in 4.4.2 shall be subjected to the test marked "X" in column 2 of table II. Failure of the test shall be cause for rejection.

4.4.3.2 Sample. Samples selected in accordance with 4.4.1 which have passed the individual test specified in 4.4.3.1 shall be subjected to the test marked "X" in column 3 of table II.

4.5 Inspection schedule.

Table I. Defects.

Defect	Requirement paragraph
101. Dimensions not as shown	3.1
102. Absence of any part or component	3.1
103. Misalinement of parts or component	3.1
104. Materials or components not as specified	3.1
105. Treatment and painting not as specified	3.13
106. Absence or incorrectness of identification markings	3.15
107. Workmanship not as specified	3.16

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Table II. Test Schedule

Initial Production	Quality conformance		Test	Test Paragraph	Requirement Paragraph
	Individual	Sample			
X	X	-	Operation	4.6.1	3.3
X	-	X	Operational reliability	4.6.2	3.8,3.14
X	-	-	Rainfall	4.6.3	3.6,3.8 3.10

4.6 Test procedures.

4.6.1 Operation. Connect the lamp assembly to a 120-volt, ac power source; plug the lamp cords and extension light into the outlet box and place the switches in the "on" position. Erect the mast assembly from its retracted position to its maximum height and lock each mast section rigid and straight. Actuate the switches on the control box from the "off" to the "on" position five times for switch control of lamp operation. Nonoperation of the lamps (with exception of normal filament burn-out) or malfunction of either electrical switch or parts of the mast shall constitute failure of this test.

4.6.2 Operational reliability. Test the lamp assembly in accordance with 4.6.1 except that the mast shall be extended and collapsed 100 times and the electrical switches shall be actuated from the "off" to the "on" position 100 times. Apply 60 inch-pounds of torque to a lamp installed in one of the sockets. Nonconformance to 3.8 or 3.14 shall constitute failure of this test.

4.6.3 Rainfall. With the mast erected to normal operation position and the projector lamps operating, subject the lamp assembly to a simulated rainfall of 4 inches plus or minus 1 inch per hour for 1 hour. After exposure to the rain, examine the lamp assembly. Evidence of water penetration into the electrical components or malfunctioning of the lamps shall constituee failure of this test.

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4.7 Inspection of preparation for delivery.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Containers not as specified for level A or B. Each incorrect container shall be considered one defect.
109.	Strapping not zinc coated for level A.
110.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The carrying case, containing the components comprising a complete floodlight set secured in place as specified in 3.12 shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style I. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.1.2 Level B. The carrying case, containing the components comprising a complete floodlight set secured in place as specified in 3.12 shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.1.3 Level C. The carrying case, containing the components comprising a complete floodlight set secured in place as specified in 3.12 shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. If the carrying case is not individually packed in a shipping container the carrying case lid shall be secured in a manner to prevent accidental opening or pilferage of contents.

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5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lamp assembly covered by this specification is intended to illuminate aircraft for maintenance operations in exposed locations under all weather conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced lamp assembly (see 3.2).
(c) Color of paint (see 3.13).
(d) Level of packing required (see 5.1).

6.3 First produced lamp assembly. Any changes or deviations of production lamp assemblies from the approved first produced lamp assembly during production will be subject to approval of the contracting officer. Approval of the first produced lamp assembly will not relieve the supplier of his obligation to furnish lamp assemblies conforming to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Review interest:

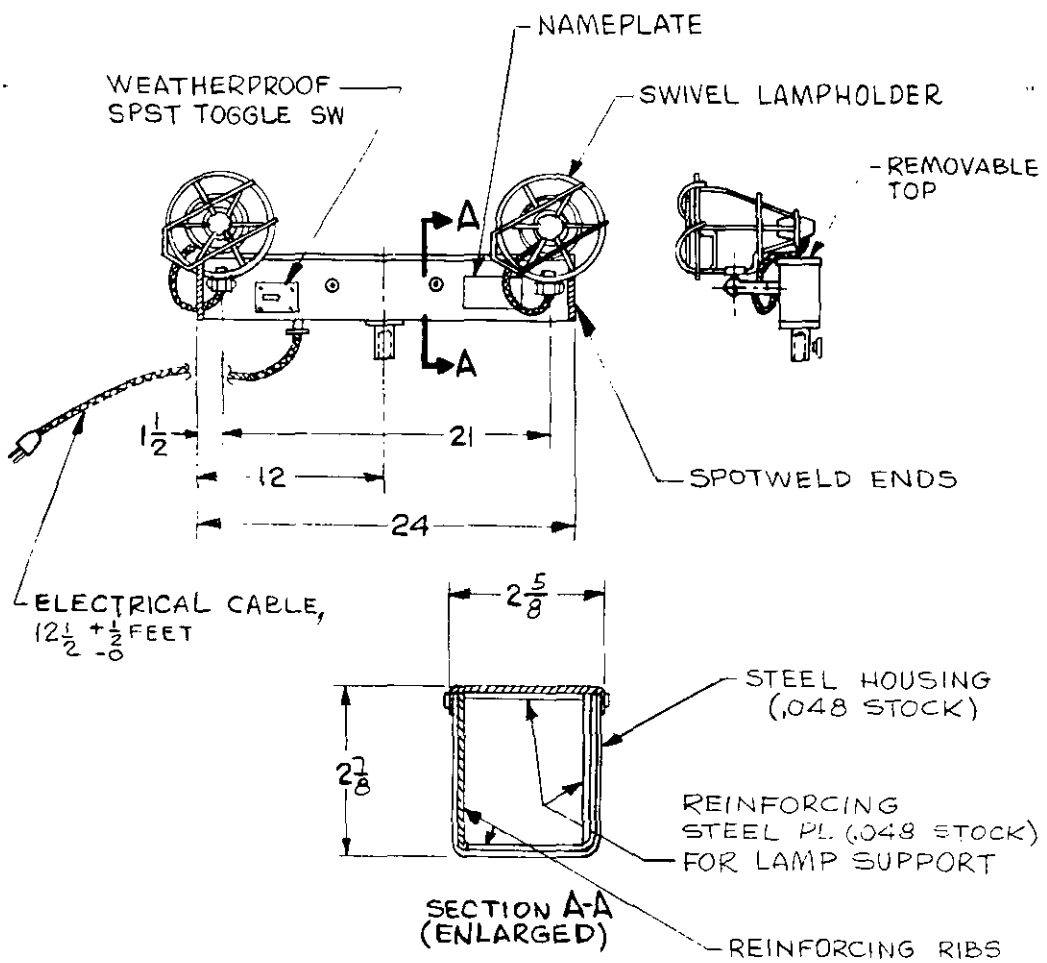
Army - SM, MD

User interest:

Army - EL

Project No. 6230-A149

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NOTES:

1. DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
2. TOLERANCES : ± 1/16
3. SCALE : NONE

FIGURE 1.
LAMP-HEAD ASSEMBLY

X-2472

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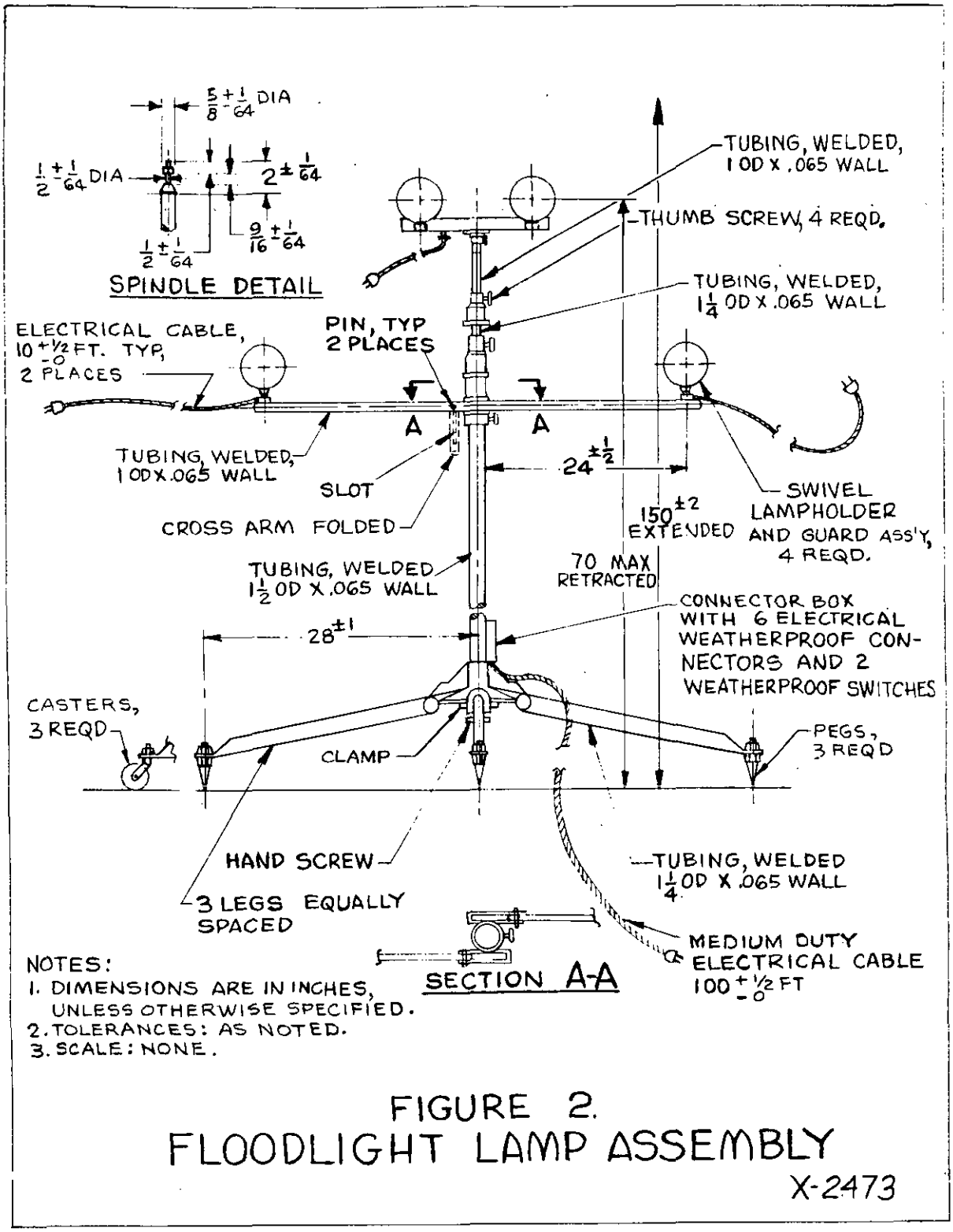
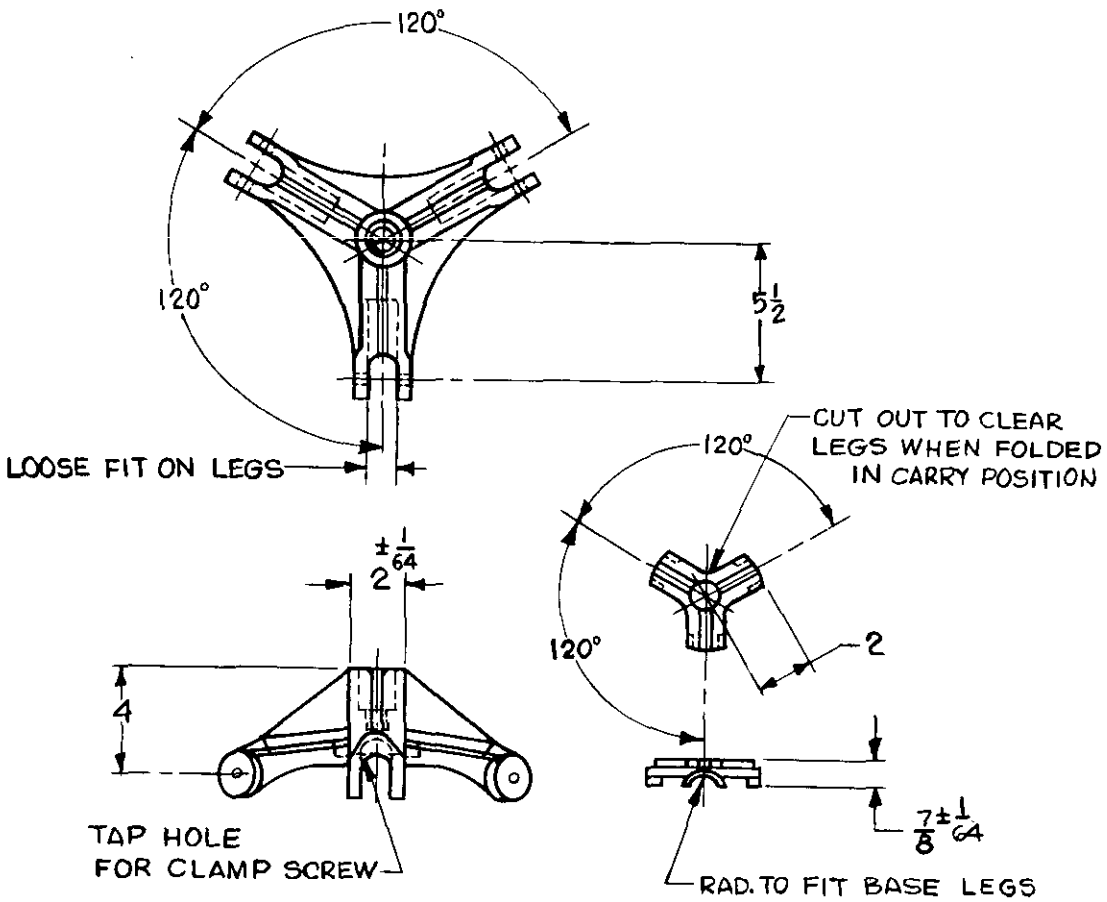


FIGURE 2.
FLOODLIGHT LAMP ASSEMBLY
X-2473

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NOTES:

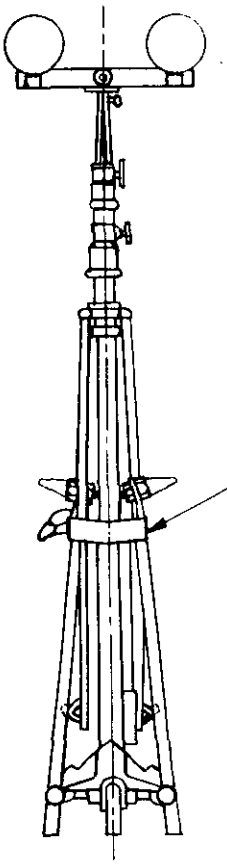
1. DIMENSIONS ARE IN INCHES.
2. TOLERANCES: FRACTIONS ± 1/8, ANGLES ± 1°.
3. SCALE: NONE.
4. MATL: MALLEABLE IRON CASTING.

FIGURE 3.

BASE CASTING AND LEG CLAMP

X-2474

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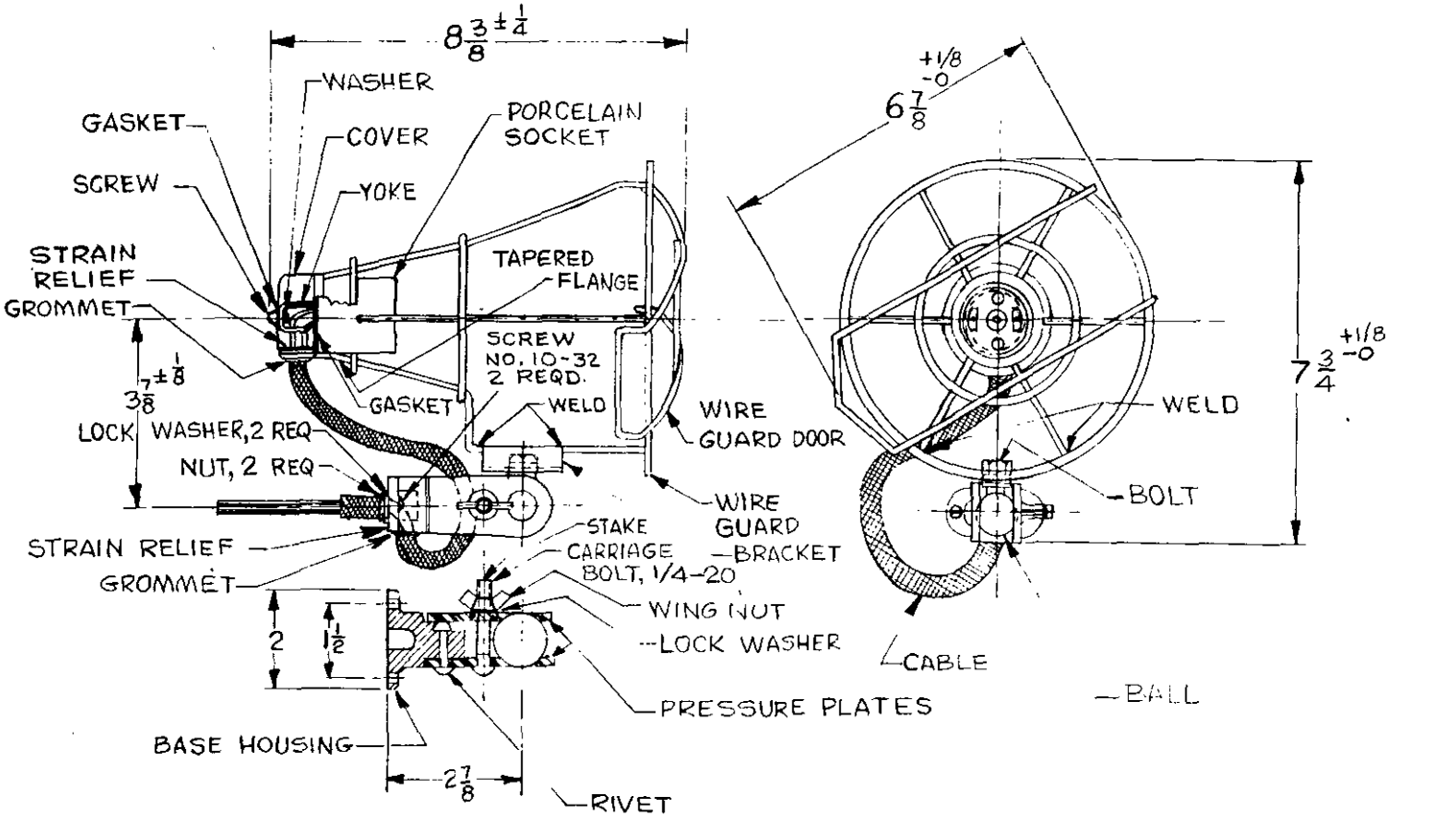
WEB STRAP, $1\frac{1}{2}$ WIDE, WITH ADJUSTER BUCKLE, ATTACHED TO CROSS ARM, FOR STRAPPING MAST IN CARRYING POSITION.

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. SCALE : NONE.

FIGURE 4.
MAST IN CARRYING POSITION.

X-2475A

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NOTES:

1. DIMENSIONS ARE IN INCHES.
2. TOLERANCES: ± 1/64
3. SCALE: NONE.

FIGURE 5.

SWIVEL LAMPHOLDER AND GUARD ASSEMBLY.

X-24176A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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